

1 Jensen Inequality

Recall: Recall that for a function

$$f : I \longrightarrow \mathbb{R},$$

where I is an interval on $\mathbb{R}$. If for $\forall x_1, x_2 \in I$ and $\forall \lambda \in [0, 1]$, we have

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2),$$

then we say that f is a **convex function**. If for $\forall x_1, x_2 \in I$ and $\forall \lambda \in [0, 1]$, we have

$$f(\lambda x_1 + (1 - \lambda)x_2) < \lambda f(x_1) + (1 - \lambda)f(x_2),$$

then we say that f is a **strictly convex function**.

If for $\forall x_1, x_2 \in I$ and $\forall \lambda \in [0, 1]$, we have

$$f(\lambda x_1 + (1 - \lambda)x_2) \geq \lambda f(x_1) + (1 - \lambda)f(x_2),$$

then we say that f is a **concave function**. If for $\forall x_1, x_2 \in I$ and $\forall \lambda \in [0, 1]$, we have

$$f(\lambda x_1 + (1 - \lambda)x_2) > \lambda f(x_1) + (1 - \lambda)f(x_2),$$

then we say that f is a **strictly concave function**.

Recall: Recall that a second-order differentiable real function f defined on an interval I on $\mathbb{R}$ is convex (concave) $\iff f''(x) > 0$ (< 0) for $\forall x \in I$.

Proposition 1. (Jesen Inequality) Let $f : I \longrightarrow \mathbb{R}$ be a strictly convex function defined on an interval I on $\mathbb{R}$, $x_i \in I$ ($i = 1, 2, \dots, n$), let $\{\lambda_i\}_{i=1}^n \subseteq [0, 1]$ such that $\sum_{i=1}^n \lambda_i = 1$, then we have

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i).$$

Similarly, if f is a strictly concave function, we have

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \geq \sum_{i=1}^n \lambda_i f(x_i).$$

Moreover, the "=" in the two inequalities hold $\iff x_1 = x_2 = \dots = x_n$.

Proof: By induction on n .

1° When $n = 1$, it's clear that $f(x_1) \leq f(x_1)$.

2° Suppose the conclusion holds for all $n \leq k$.

3° Now, let $n = k + 1$, in this case, we have $\sum_{i=1}^{k+1} \lambda_i = 1$, denote

$$\Lambda := \sum_{i=1}^k \lambda_i.$$

If $\Lambda = 0$, it degenerates to the case 1°. So we only need to consider the case that $\Lambda > 0$, note that $\Lambda + \lambda_{k+1} = 1$. Now we denote the point

$$y := \sum_{i=1}^k \frac{\lambda_i}{\Lambda} x_i,$$

so by the definition of convex functions, we see that

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) = f(\Lambda y + \lambda_{k+1} x_{k+1}) \leq \Lambda f(y) + \lambda_{k+1} f(x_{k+1}).$$

Since $\sum_{i=1}^k \frac{\lambda_i}{\Lambda} = 1$, so by the induction hypothesis, we have

$$f(y) = f\left(\sum_{i=1}^k \frac{\lambda_i}{\Lambda} x_i\right) \leq \sum_{i=1}^k \frac{\lambda_i}{\Lambda} f(x_i),$$

i.e., we get

$$f\left(\sum_{i=1}^{k+1} \lambda_i x_i\right) \leq \Lambda \cdot \sum_{i=1}^k \frac{\lambda_i}{\Lambda} f(x_i) + \lambda_{k+1} f(x_{k+1}) = \sum_{i=1}^{k+1} \lambda_i f(x_i).$$

By the induction on n , it's easy to see that the two "=" hold iff $x_1 = x_2 = \dots = x_n$ for strictly convex (concave) function f . ■

Corollary 1. Let $f : I \rightarrow \mathbb{R}$ be a convex function defined on an interval I on $\mathbb{R}$, $x_i \in I$ ($i = 1, 2, \dots, n$), then we have

$$f\left(\frac{\sum_{i=1}^n x_i}{n}\right) \leq \frac{\sum_{i=1}^n f(x_i)}{n}.$$

Similarly, if f is a concave function, then

$$f\left(\frac{\sum_{i=1}^n x_i}{n}\right) \geq \frac{\sum_{i=1}^n f(x_i)}{n}.$$

Proof: Take $\lambda_i = \frac{1}{n}$ for $i = 1, 2, \dots, n$. ■

Example 1. Let $f(x) := x^2$ defined on $\mathbb{R}$, so $f''(x) = 2$ for $\forall x \in \mathbb{R}$, i.e., $f(x)$ is convex on $\mathbb{R}$, so by Corollary 1, we see that

$$f\left(\frac{\sum_{i=1}^n x_i}{n}\right) = \left(\frac{\sum_{i=1}^n x_i}{n}\right)^2 \leq \frac{\sum_{i=1}^n x_i^2}{n} = \frac{\sum_{i=1}^n f(x_i)}{n},$$

i.e.,

$$\frac{\sum_{i=1}^n x_i}{n} \leq \sqrt{\frac{\sum_{i=1}^n x_i^2}{n}} \quad (\forall x_i \in \mathbb{R}).$$

Example 2. (The HM-GM-AM-QM Inequality) Let $x_i > 0$ ($i = 1, 2, \dots, n$) be any given real numbers, then we have

$$\frac{n}{\sum_{i=1}^n \frac{1}{x_i}} \leq \sqrt[n]{\prod_{i=1}^n x_i} \leq \frac{\sum_{i=1}^n x_i}{n} \leq \sqrt{\frac{\sum_{i=1}^n x_i^2}{n}}.$$

Proof: Define the function

$$f(x) := \ln x,$$

then we see that

$$f''(x) = -\frac{1}{x^2} < 0 \quad \text{for } \forall x \in (0, +\infty),$$

which means $f(x) = \ln x$ is a concave function. So by the Jensen Inequality (Proposition 1), we see that

$$f\left(\sum_{i=1}^n \frac{x_i}{n}\right) \geq \sum_{i=1}^n \frac{1}{n} f(x_i),$$

i.e.,

$$\ln\left(\sum_{i=1}^n \frac{x_i}{n}\right) \geq \sum_{i=1}^n \frac{1}{n} \ln(x_i) = \frac{1}{n} \sum_{i=1}^n \ln(x_i) = \ln\left(\left(\prod_{i=1}^n x_i\right)^{\frac{1}{n}}\right).$$

Note that $f(x) = \ln x$ is a strictly monotonic function on $(0, +\infty)$. So we immediately get

$$\sum_{i=1}^n \frac{x_i}{n} \geq \left(\prod_{i=1}^n x_i\right)^{\frac{1}{n}},$$

i.e.,

$$\sqrt[n]{\prod_{i=1}^n x_i} \leq \frac{\sum_{i=1}^n x_i}{n};$$

Now, for $x_i \in (0, +\infty)$, denote

$$y_i := \frac{1}{x_i} \in (0, +\infty),$$

so we have

$$\left(\prod_{i=1}^n y_i\right)^{\frac{1}{n}} \leq \frac{\sum_{i=1}^n y_i}{n},$$

which means that

$$0 < \left(\prod_{i=1}^n \frac{1}{x_i}\right)^{\frac{1}{n}} = \frac{1}{\left(\prod_{i=1}^n x_i\right)^{\frac{1}{n}}} \leq \frac{\sum_{i=1}^n \frac{1}{x_i}}{n},$$

which implies that

$$\frac{n}{\sum_{i=1}^n \frac{1}{x_i}} \leq \sqrt[n]{\prod_{i=1}^n x_i};$$

The last inequality

$$\frac{\sum_{i=1}^n x_i}{n} \leq \sqrt{\frac{\sum_{i=1}^n x_i^2}{n}}$$

is directly from the Example 1. ■

Remark: For $x_i \in (0, +\infty)$, we write

$$\text{HM} \triangleq \text{harmonic mean} := \frac{n}{\sum_{i=1}^n \frac{1}{x_i}},$$

$$\text{GM} \triangleq \text{geometric mean} := \sqrt[n]{\prod_{i=1}^n x_i},$$

$$\text{AM} \triangleq \text{arithmetic mean} := \frac{\sum_{i=1}^n x_i}{n},$$

$$\text{QM} \triangleq \text{quadratic mean} := \sqrt{\frac{\sum_{i=1}^n x_i^2}{n}},$$

hence the chain of inequalities in the Example 2 above can be shortly written as

$$\text{HM} \leq \text{GM} \leq \text{AM} \leq \text{QM}.$$

Corollary 2. Let $a, b \in (0, +\infty)$, then we have

$$\frac{2}{\frac{1}{a} + \frac{1}{b}} \leq \sqrt{ab} \leq \frac{a+b}{2} \leq \sqrt{\frac{a^2 + b^2}{2}}.$$

Proof: Directly comes from Example 2. ■

2 Hölder's Inequality

Lemma 1. (Young's Inequality) Let $p, q \in (1, +\infty)$ such that

$$\frac{1}{p} + \frac{1}{q} = 1,$$

then for $\forall a, b \in (0, +\infty)$, we have

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

where the equality " = " holds $\iff a^p = b^q$.

Proof: Define the function

$$f(x) := e^x,$$

then we see that

$$f''(x) = e^x > 0 \text{ for } \forall x \in \mathbb{R},$$

which means that $f(x) = e^x$ is a convex function. Now, we take $x = p \ln a$ and take $y = q \ln b$, then by the Jensen Inequality (Proposition 1), we get

$$f\left(\frac{1}{p} \cdot (p \ln a) + \frac{1}{q} \cdot (q \ln b)\right) = f\left(\frac{1}{p}x + \frac{1}{q}y\right) \leq \frac{1}{p}f(x) + \frac{1}{q}f(y),$$

i.e.,

$$e^{\frac{1}{p}(p \ln a) + \frac{1}{q}(q \ln b)} = e^{\ln a + \ln b} = e^{\ln(ab)} = ab \leq \frac{1}{p}e^{p \ln a} + \frac{1}{q}e^{q \ln b} = \frac{1}{p}(e^{\ln a})^p + \frac{1}{q}(e^{\ln b})^q = \frac{a^p}{p} + \frac{b^q}{q},$$

"=" holds $\iff p \ln a = q \ln b$, i.e., $a^p = b^q$. ■

By using the Young's Inequality, we can get the following Hölder's Inequality.

Theorem 1. (Hölder's Inequality) Let $\{a_i\}_{i=1}^n \subseteq (0, +\infty)$ and let $\{b_i\}_{i=1}^n \subseteq (0, +\infty)$, suppose $p, q \in (1, +\infty)$ such that

$$\frac{1}{p} + \frac{1}{q} = 1,$$

then we have

$$\sum_{k=1}^n a_k b_k \leq \left(\sum_{k=1}^n a_k^p \right)^{\frac{1}{p}} \left(\sum_{k=1}^n b_k^q \right)^{\frac{1}{q}},$$

where the equality "=" holds $\iff$ there exists a constant $\lambda \in \mathbb{R}$ such that $a_k^p = \lambda b_k^q$ for $\forall 1 \leq k \leq n$.

Proof: Denote

$$A := \left(\sum_{i=1}^n a_i^p \right)^{\frac{1}{p}}, \quad B := \left(\sum_{i=1}^n b_i^q \right)^{\frac{1}{q}}.$$

We are going to prove that $\sum_{k=1}^n a_k b_k \leq AB$:

Denote

$$x_k := \frac{a_k}{A} = \frac{a_k}{\left(\sum_{i=1}^n a_i^p \right)^{\frac{1}{p}}}, \quad y_k := \frac{b_k}{B} = \frac{b_k}{\left(\sum_{i=1}^n b_i^q \right)^{\frac{1}{q}}}.$$

So we have

$$\sum_{k=1}^n x_k^p = \sum_{k=1}^n \left(\frac{a_k}{\left(\sum_{i=1}^n a_i^p \right)^{\frac{1}{p}}} \right)^p = 1, \quad \sum_{k=1}^n y_k^q = \sum_{k=1}^n \left(\frac{b_k}{\left(\sum_{i=1}^n b_i^q \right)^{\frac{1}{q}}} \right)^q = 1.$$

By the Young's Inequality, we get

$$\frac{a_k b_k}{AB} = x_k y_k \leq \frac{x_k^p}{p} + \frac{y_k^q}{q} = \frac{1}{p} \cdot \frac{a_k^p}{A^p} + \frac{1}{q} \cdot \frac{b_k^q}{B^q} \quad \text{for every } k \in \{1, 2, \dots, n\},$$

i.e.,

$$a_k b_k \leq AB \left(\frac{1}{p} \cdot \frac{a_k^p}{A^p} + \frac{1}{q} \cdot \frac{b_k^q}{B^q} \right) \quad \text{for every } k \in \{1, 2, \dots, n\}.$$

Thus

$$\begin{aligned} \sum_{k=1}^n a_k b_k &\leq \sum_{k=1}^n \left(AB \left(\frac{1}{p} \cdot \frac{a_k^p}{A^p} + \frac{1}{q} \cdot \frac{b_k^q}{B^q} \right) \right) \\ &= AB \left(\frac{1}{p} \cdot \frac{1}{A^p} \cdot \sum_{k=1}^n a_k^p + \frac{1}{q} \cdot \frac{1}{B^q} \cdot \sum_{k=1}^n b_k^q \right) \\ &= AB \left(\frac{1}{p} \cdot \frac{1}{A^p} \cdot A^p + \frac{1}{q} \cdot \frac{1}{B^q} \cdot B^q \right) \\ &= AB \left(\frac{1}{p} + \frac{1}{q} \right) \\ &= AB. \end{aligned}$$

Hence we get the desired inequality. Note that the "=" holds $\iff x_k y_k = \frac{x_k^p}{p} + \frac{y_k^q}{q}$ for $\forall k \in \{1, 2, \dots, n\}$, so by Young's Inequality, we know that "=" holds $\iff x_k^p = y_k^q$ ($\forall 1 \leq k \leq n$). Thus

$$"=" \text{ holds } \iff \frac{a_k^p}{A^p} = \frac{b_k^q}{B^q} \iff \frac{a_k^p}{b_k^q} = \frac{A^p}{B^q} \triangleq \lambda,$$

which completes the proof. ■

Corollary 3. (Cauchy-Schwarz Inequality) Let $\{a_i\}_{i=1}^n \subseteq (0, +\infty)$ and let $\{b_i\}_{i=1}^n \subseteq (0, +\infty)$, then we have

$$\left(\sum_{k=1}^n a_k b_k \right)^2 \leq \left(\sum_{k=1}^n a_k^2 \right) \left(\sum_{k=1}^n b_k^2 \right).$$

Proof: Take $p = q = 2$ in the Hölder's Inequality. ■

Proposition 2. Let $\{a_i\}_{i=1}^n \subseteq (0, +\infty)$ and let $\{b_i\}_{i=1}^n \subseteq (0, +\infty)$, then for any $m \in \mathbb{N}$, we have

$$\sum_{i=1}^n \frac{a_i^{m+1}}{b_i^m} \geq \frac{(\sum_{i=1}^n a_i)^{m+1}}{(\sum_{i=1}^n b_i)^m},$$

where "=" holds $\iff \frac{a_1}{b_1} = \frac{a_2}{b_2} = \dots = \frac{a_n}{b_n}$.

Proof: For any given $m \in \mathbb{N}$, define

$$p_1 := m + 1, \quad p_2 := \frac{m + 1}{m},$$

hence $p_1, p_2 \in (1, +\infty)$ and we have

$$\frac{1}{p_1} + \frac{1}{p_2} = 1.$$

If we denote

$$x_i := \left(\frac{a_i^{m+1}}{b_i^m} \right)^{\frac{1}{p_1}} = \left(\frac{a_i^{m+1}}{b_i^m} \right)^{\frac{1}{m+1}}, \quad y_i := b_i^{\frac{1}{p_2}} = b_i^{\frac{m}{m+1}},$$

then by Hölder's Inequality, we get

$$\sum_{i=1}^n x_i y_i \leq \left(\sum_{i=1}^n x_i^{p_1} \right)^{\frac{1}{p_1}} \left(\sum_{i=1}^n y_i^{p_2} \right)^{\frac{1}{p_2}},$$

since

$$\text{LHS} = \sum_{i=1}^n \left(\left(\frac{a_i^{m+1}}{b_i^m} \right)^{\frac{1}{m+1}} \cdot b_i^{\frac{m}{m+1}} \right) = \sum_{i=1}^n a_i,$$

$$\text{RHS} = \left(\sum_{i=1}^n \frac{a_i^{m+1}}{b_i^m} \right)^{\frac{1}{m+1}} \cdot \left(\sum_{i=1}^n b_i \right)^{\frac{m}{m+1}},$$

so these lead to

$$\sum_{i=1}^n a_i \leq \left(\sum_{i=1}^n \frac{a_i^{m+1}}{b_i^m} \right)^{\frac{1}{m+1}} \cdot \left(\sum_{i=1}^n b_i \right)^{\frac{m}{m+1}},$$

hence we get

$$\left(\sum_{i=1}^n a_i\right)^{m+1} \leq \left(\sum_{i=1}^n \frac{a_i^{m+1}}{b_i^m}\right) \cdot \left(\sum_{i=1}^n b_i\right)^m,$$

i.e.,

$$\sum_{i=1}^n \frac{a_i^{m+1}}{b_i^m} \geq \frac{(\sum_{i=1}^n a_i)^{m+1}}{(\sum_{i=1}^n b_i)^m},$$

the equality "=" holds $\iff$ there is a constant $\lambda \in \mathbb{R}$ such that

$$\frac{x_i^{p_1}}{y_i^{p_2}} = \lambda,$$

for all $1 \leq i \leq n$, which means

$$\frac{a_i}{b_i} = \lambda^{\frac{1}{m+1}} \triangleq k \quad \text{for } \forall 1 \leq i \leq n,$$

which completes the proof. ■

Corollary 4. Let $\{a_i\}_{i=1}^n \subseteq (0, +\infty)$ and let $\{b_i\}_{i=1}^n \subseteq (0, +\infty)$, then we have

$$\sum_{i=1}^n \frac{a_i^2}{b_i} \geq \frac{(\sum_{i=1}^n a_i)^2}{\sum_{i=1}^n b_i},$$

where the equality "=" holds $\iff \frac{a_i}{b_i} = \frac{a_j}{b_j}$ ($\forall 1 \leq i, j \leq n$). In particular, if $a, b, x, y \in (0, +\infty)$, then we have

$$\frac{x^2}{a} + \frac{y^2}{b} \geq \frac{(x+y)^2}{a+b},$$

where the equality "=" holds $\iff \frac{x}{a} = \frac{y}{b}$.

Proof: Take $m = 1$ in the proposition above, we immediately get this. ■